

Graduation Project Steps

Step 0: Objective and Methodology

Objective:

To compare Egypt's electric power generation and renewable energy transition with nearby or similar countries in terms of production, efficiency, and sustainability.

Methodology:

- Identify countries with total electricity generation within $\pm 10\%$ of Egypt's level.
- Analyze similarities in generation sources (fossil, hydro, solar, wind, nuclear, etc.).
- Collect data from reliable sources such as the International Energy Agency (IEA), World Bank, and Our World in Data.
- Use tools such as Excel, Python, and Power BI to clean, process, and visualize data.

Step 1: Nearest Countries in Electricity Generation

Identify the nearest countries to Egypt in terms of electric power generation (within $\pm 10\%$). The output should be a list of countries whose total generation closely matches Egypt's electricity generation.

Output:

List of the nearest countries ($\pm 10\%$) to Egypt with respect to the total amount of electricity generation. (Presented in a Dashboard)

Step 2: Sources of Generation

Identify the sources of electricity generation for Egypt and for the selected countries (e.g., fossil fuels, hydro, solar, wind, nuclear). Determine which countries have a similar mix of energy sources.

If no close matches are found, use regional comparisons:

- Africa region
- MENA region

Output:

Dashboard showing comparison of generation sources for Egypt and the selected countries.

Step 3: Energy Performance Indicators

For each country identified in Steps 1 & 2, collect the following data from the IEA website and visualize them in dashboards:

- Emissions: Carbon and greenhouse gas emissions related to power generation.
- Efficiency and Demand: Energy conversion efficiency and total electricity demand.
- Renewables: Share and growth rate of renewable energy in the total mix.

Output:

Dashboards displaying emissions, efficiency, and renewable trends.

Step 4: Renewable Energy Focus

For each country in the comparison list, analyze the renewable energy situation in more detail:

- Potential of Renewable Energy: Solar radiation, wind speed, hydro capacity, etc.
- Vision and Goals: National strategies for renewable energy transition.
- Economic, Social, and Political Impact: How renewable energy affects employment, GDP, and stability.

Output:

Dashboard and summary of renewable energy potential, strategies, and impacts.

Step 5: Data Analysis and Visualization

Perform analytical comparisons between Egypt and the selected countries:

- Create comparative charts for electricity mix, emissions, and renewable energy growth.
- Show efficiency vs. demand relationships.
- Highlight Egypt's ranking among selected countries.

Output:

Interactive Power BI dashboard comparing performance indicators.

Step 6: Insights and Interpretation

- Identify which countries are most similar to Egypt and explain why.
- Discuss strengths and weaknesses of Egypt's current energy mix.
- Highlight key challenges and opportunities for renewable transition in Egypt.

Step 7: Recommendations

Provide clear, actionable recommendations based on findings:

- Suggest policy improvements inspired by successful countries.
- Identify promising renewable sources for Egypt (e.g., solar and wind).
- Highlight potential partnerships or investments for accelerating renewable adoption.

Step 8: Conclusion and Future Work

Conclusion:

Summarize the overall findings of the analysis, showing Egypt's current position and progress compared to its peers.

Future Work:

- Extend the analysis to include energy storage, cost analysis, and grid modernization.
- Monitor future trends and updates in renewable energy policies and technologies.